

Quang-Vinh Dang
British University Vietnam
Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

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Dear Editors,

I am pleased to submit my manuscript, “**Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces**,” for consideration as a research article in *Computers & Security*.

Large language model agents that call external tools now read email, browse the web and move money on a user’s behalf. Because they ingest untrusted content and fold it into the context that drives their next action, they are exposed to indirect prompt injection. Runtime monitors that watch only the black-box execution trace need no model access and no retraining, but the literature has converged almost entirely on one kind of evidence: whether untrusted text *looks like an instruction*. This paper asks what else the trace affords, how heterogeneous evidence should be combined, and where the information in a trace runs out.

The work makes four contributions. First, it introduces **SENTRY-Fuse**, a monitor that converts heterogeneous trace signals—one scoring the observation stream, two auditing the action stream—into conformal tail surprisals against a benign calibration split and fuses them by addition. The construction is scale-free, non-decreasing in every component and free of tuned signal weights, and the paper proves that the natural alternative of subtracting the benign training maximum is many-to-one and caps AUROC at one half. That is not hypothetical: an earlier, unpublished draft of this work did exactly that, at a cost of 0.25 AUROC and a confidently wrong explanation of the resulting failure. Second, action-side evidence is shown to be *complementary* rather than redundant: over 90 genuinely compromised trajectories the fusion reaches $\text{AUROC} = 0.957 \pm 0.004$ and a true-positive rate of 0.919 ± 0.007 at a realised 3.4% false-positive rate, against 0.944 and 0.729 for the instruction signal alone, and it cuts that signal’s across-split standard deviation from ± 0.364 to ± 0.007 . Third, it formalises the **detectability ceiling** of trace-only monitoring and then measures it: on AgentDojo the fraction of compromises leaving no trace evidence lies between 0 and 0.044 depending on how visibility is defined, putting the bound at or above 0.958, because the benchmark delivers payloads through tool outputs a successful injection must have read. The ceiling is an artefact of benchmark construction rather than a limit of the problem—a negative result about the field’s evaluation practice, and the finding I expect to be most useful to others. It also led me to withdraw a claim an earlier draft made, that a published figure exceeded this corpus’s information limit; that was an artefact of the measurement convention. Fourth, it documents a label-polarity pitfall in a widely used harness that inverts the compromised and resisted classes, with the ground-truth procedure that resolves it.

I should be explicit that the paper does *not* claim to beat the state of the art. The strongest reported trace-only figure (95.75% true positives at 3.66% false positives) comes from a preprint version that later versions withdrew, was measured on a different corpus with a different agent, and I re-implemented no baseline on my own corpus. My numbers sit in the same regime, and I say so rather than constructing a comparison the evidence does not support.

I believe the work suits *Computers & Security* in particular. It is a security paper first: it defines a threat model, evaluates a defence against real attacks on real agent traces, and reports its limits rather than only its successes. It offers two things of direct use to others: a measurement showing that the benchmarks the field relies on cannot exhibit the hardest case, and a methodological warning about how results on a standard benchmark should be read. Its negative results are reported in full—four signals I implemented and rejected, two of which an earlier draft presented as part of the method, and the finding that a hand-picked threshold outperformed the anytime-valid machinery from which the project began.

All code, the trajectory corpora, the cached model-judge scores and the analysis scripts that regenerate every number, table and figure are publicly released, and the evaluation reproduces offline without access to any model API.

I confirm that this manuscript is original, has not been published previously and is not under consideration elsewhere. There are no competing interests and no external funding was received; as sole author I am responsible for all aspects of the work. Thank you for your consideration, and I look forward to the reviewers’ comments.

Sincerely,

Quang-Vinh Dang